

Gender Diversity of Graphics Conference Leadership

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ABSTRACT

Non-male identifying researchers, including women and non-binary individuals, are underrepresented in leadership roles within computer graphics conferences. Analyzing data on these roles over time across key conferences highlights this disparity, while also identifying positive trends and worrisome constants. The goal of the following data collection is to provide grounded statistics to inform future decision making. Recognizing even small successes from the past may inspire and accelerate future improvement. The academic community’s recent failure to sufficiently address sexual harassment and malevolent behavior from members of our community (e.g. [Obr 2023]) is an urgent reminder to improve.

This is admittedly arm-chair statistics. Anyone with the internet could collect this data and create the following statistics and plots. Nevertheless, I would like to be thorough about the methodology, largely replicating Graesser et al. [2021], who examined robotics.

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1 METHODOLOGY

The most fraught aspect of this kind of data collection is the categorization of other people’s gender. I will purposefully refer to *perceived* gender to emphasize that this is *not* the result of self-reporting, which would have been better [Dodik et al. 2022]. Gender is not binary. The primary goal of this data collection is to emphasize the enduring lack of non-male people in leadership roles at disproportionate rate compared to: society at large and some computer science undergraduate populations (e.g., $\approx 49\%$ women at CMU [Frieze and Quesenberry 2019]), though perhaps not computer science faculty populations (e.g., $\approx 24\%$ of budgetary tenure-stream computer science faculty at University of Toronto) or computer graphics faculty populations (e.g., $\approx 15\%$ of PhD-supervising, *SIGGRAPH*-publishing faculty in Canada).

Here, gender is categorized according to pronoun use. Only she/her (female) and he/him (male) pronouns were encountered. Pronoun categorization is imperfect: pronouns may be out of date,

inaccurate, or insufficiently expressive. I’ll frequently refer to “non-male.” This is not to imply a male default, but rather to emphasize the disproportionate under-representation of all other groups relative to the dominant male group.

Each data point is the perceived gender of a person holding a leadership role at a graphics conference. The data is collected across a matrix of variables: conference $\times$ role $\times$ year. Six conferences are considered: *SIGGRAPH North America (SNA)* and *SIGGRAPH Asia (SA)*, the premier graphics conferences; *Eurographics (EG)* and *Pacific Graphics (PG)*, leading provincial general-topic conferences; and *Symposium on Geometry Processing (SGP)* and *Symposium on Computer Animation (SCA)*, well-respected specialized conferences.

Three conference leadership roles are considered, collecting the typically most visible or high-profile roles of a conference: *Conference Chair*, also referred to as “general chair” (typically one or two per conference); *Papers Chair*, also referred to as “program chair” (typically one or two); and *Keynote Speaker*, also referred to as “invited”, “featured” or “plenary” (typically one to five).

Limiting to these few high-profile roles prevents saturation and diffusion of other – albeit interesting and important – roles (e.g., posters chair, technical papers committee, paper authors). The selected roles also carry fairly consistent prestige relative to their respective conferences. I specifically do not include “local organizing chairs” when a conference chair exists. Similarly, I do not include “invited speakers” when higher-billed keynote speakers exist. For recent *SIGGRAPHs*, I *do* include the Conference Papers Co-Chairs. Importantly, the selected leadership roles are often accompanied with website biographies revealing gendered pronouns.

The years 2010–2023 are considered. Every conference occurred in each year to some degree (many remotely during the COVID-19 global emergency; *Pacific Graphics* 2020 papers presented at the 2021 conference). This timespan reaches back far enough to observe multiple years in a row of multiple conferences with only perceived-male people taking on the leadership roles.

Past conference websites are woefully under-maintained, many only available through The Wayback Machine.

2 DATA SUMMARY

Complete data is available as a published Google Sheet, which I link to and discuss more details in a companion blog article¹. The following bar visualization widths are proportional to sample size. Across the 543 total positions, we record only 19% non-male.

Across the six conferences totalling positions over all roles, non-male representation ranges from 12% (SA) to 24% (EG). As SNA

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¹<https://alecjacobson.com/weblog/gender-diversity-in-graphics-conference-leadership.html>

stands out with the largest percentage non-male, let's scratch further. At that conference, across roles we see that the conference chair diversity clearly helps raise up the total.

Meanwhile, papers chairs appear to have the least gender diversity. Across conferences, non-male representation ranges from 11% (SCA) to 25% (SA).

The data collection also reveals a sense of how large (or small) our community's leadership circle is. Out of the 543 total *positions*, there appear: 386 unique people, 322 unique perceived-male people (83%), 63 unique perceived-female people (16%), 0 perceived-non-binary people (0%).

To get a sense of the seeming power-law distribution, the top 10 busiest unique names accounted for $\approx 10\%$ of all the total positions. We saw above that 19% of positions are filled by non-male people, but see within this “busiest 10” that it skews to 38% female. This supports the hypothesis that the same women are repeatedly asked to fill conference leadership roles. These roles are prestigious and recognitions of excellence, but also work; if gender diversity is achieved at the position level by tapping work from the same small group, it could be considered a diversity tax. For example, out of 162 papers chair positions, 17 perceived-female individuals account for all 26 occurrences when a non-male person filled the role.

Another way of looking at the distribution of positions to unique individuals is to take the average number of positions any one individual might have: individual males held 1.37 positions on average, while individual non-males held 1.61 positions on average.

Yet another way to look at it, is to consider the percentage of unique individuals over the total number of positions filled by a given gender category:

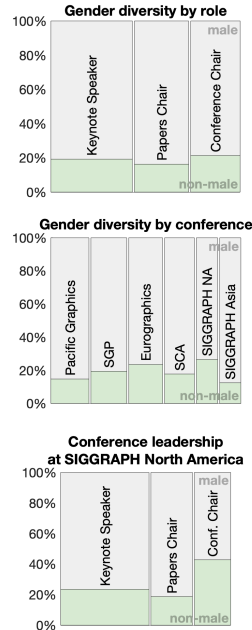
322 male individuals / 441 positions they filled = 73%,
64 female individuals / 102 positions they filled = 62%,

We can visualize percentages of non-male positions at each conference (scattered dots) and their accumulation (black line) over the years in question. Especially obscured in this visualization is the abundance of “all-male leadership” conferences (0% non-male positions across conference chairs, papers chairs, and keynote speakers).

In 2014, all six conferences failed to fill a single position with a non-male person. All-male leaderships continue as recently as *SIGGRAPH Asia 2022*.

3 OBSERVATIONS

The total line in the visualization above does appear to show a positive trend over time. In the last few years it's not clear whether that increase is continuing or stagnating.



In late 2017, the #metoo movement accumulated widespread media attention, intersecting with the graphics community directly when in early 2018 the *Wall Street Journal* reported on sexual harassment occurring at a past *SIGGRAPH* conference [Korn 2018].

In 2019, *SIGGRAPH North America* records the only “all-female” conference leadership. In that year, four conferences increased the percentage of non-male positions. All four decreased the next year.

Paper and/or conference chairs typically (help) select keynote speakers for a conference. A reasonable hypothesis is that all-male-chaired conferences are more likely to produce all-male keynote lineups. We can test the corresponding null hypothesis: “having at least one non-male chair is independent of having at least one non-male keynote speaker.” The contingency table for our data is:

	all-male keynote(s)	≥ 1 non-male keynote
all-male chair(s)	29	9
≥ 1 non-male chair	26	19

with $\chi^2 = 16.88$ and $p = 0.00004$, confidently rejecting the null hypothesis. Alternatively, we could consider whether the percentage of non-male individuals in chair positions implies a larger percentage of non-male keynoters. Indeed, we see a moderate positive linear correlation of these continuous variables with Pearson's correlation coefficient $\rho = 0.38$, $p = 0.0004$. Graesser et al. [2021] found similar correlations in robotics conferences. The causal model (chairs' gender $\rightarrow$ keynotes' gender) could of course be confounded by other variables eroding the value of these correlations as evidence. Meanwhile, the collected pronoun data may not reflect perceived or presented gender at the time of leadership selection.

4 CONCLUSIONS

To approach the diversity of our student and societal populations, the graphics community must continue to increase the number of non-male individuals in conference leadership positions. We should stop hosting conferences with all-male leaderships. By increasing the diversity of conference and paper chairs, we may also help increase the diversity of keynoters. We should train a more diverse community of researchers and put a larger number of unique individuals into leadership roles. We should also increase the number of non-binary individuals in conference leadership positions. My data's imperfect categorization muddies the comparison, but zero perceived non-binary individuals out of 386 (0%) is less than general population surveys (e.g., 1.6% in the U.S.A. [Brown 2022]).

Focusing solely on gender – as this data and discussion does – may steal much-needed attention from other gaps in representational diversity. Similarly, focusing on representational diversity may steal attention from other inclusivity problems within the graphics community. Perhaps the final conclusion is to get out of the armchair and take action.

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